

CP317 - Software Engineering

Project Title: Car Rental System

Deliverable: Sprint 2

Group ID: Team # 28

Team Members and Roles:

- SabilUllah Hussaini (Product Owner)
- Sahil Sandhu (Product Owner & Developer)
- Moosa Ahmed (Scrum Master & Developer)
- Mahad Farrukh (Developer)
- Ayesha Raheel (Developer)
- Kritika Kamath (Developer)
- Shalin Panjwani (Developer)

Sprint Goal

To implement the core rental booking functionality, user dashboard, and administrative vehicle management system, transforming our prototype into a fully functional car rental platform where users can complete the entire rental journey from search to reservation management.

Prioritization Rationale: The stories selected for Sprint 2 represent the logical progression from our Sprint 1 foundation. After establishing user authentication and vehicle search in Sprint 1, these features deliver the essential business logic that makes the system commercially viable. Users can now complete reservations and manage their bookings, while administrators can maintain the vehicle fleet. This creates a complete MVP that provides real value to both customers and rental company staff.

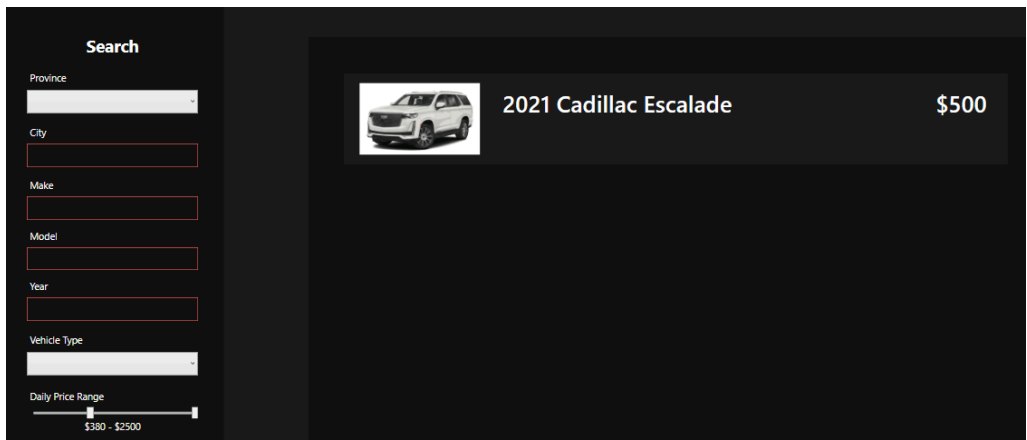
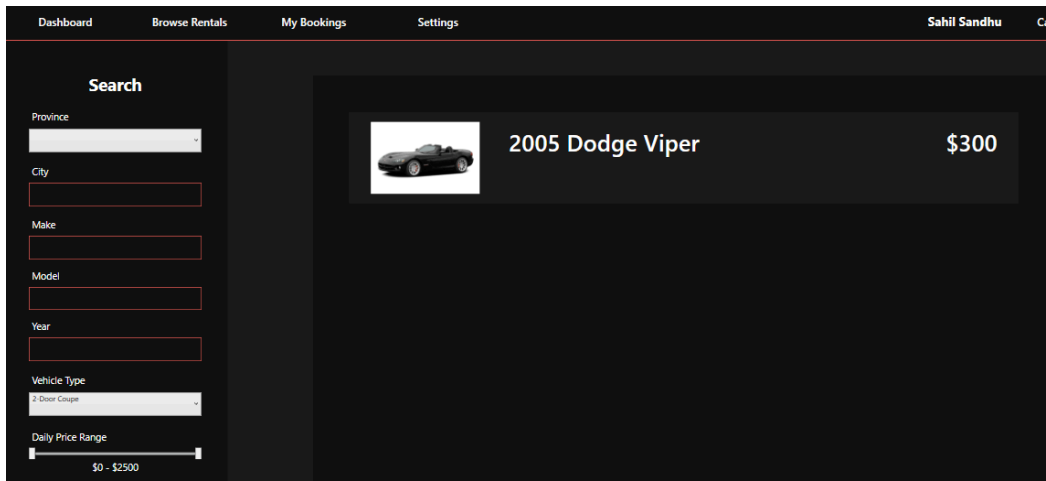
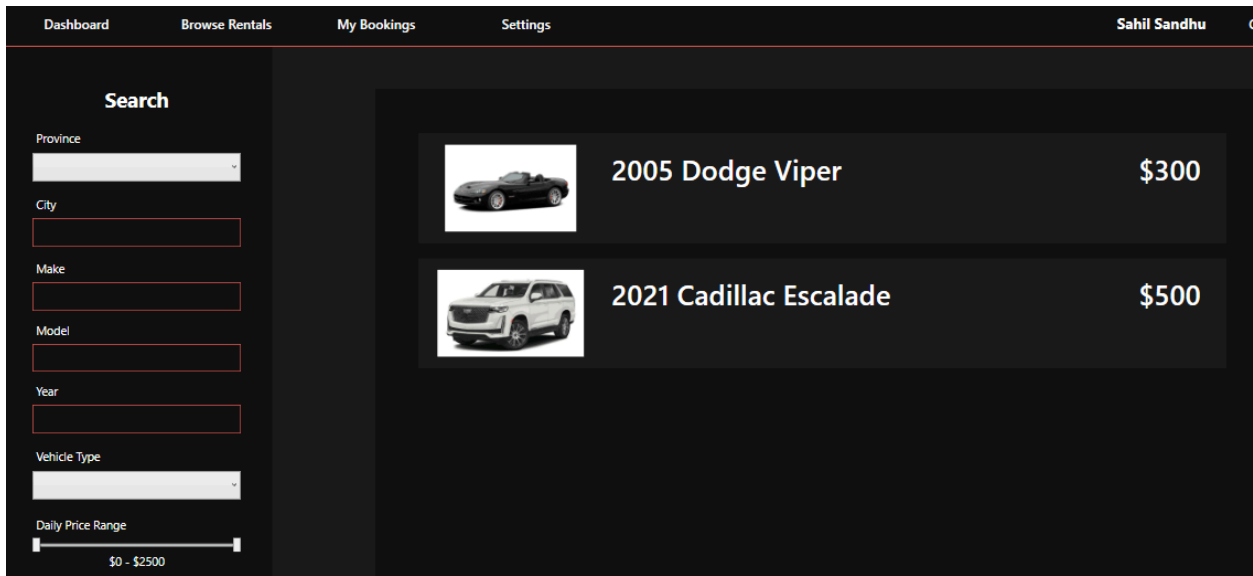
Story ID	Story Title	Story Points	Priority	Assigned Team Members	Status	Key Tasks
USER-1	Personal Dashboard for Reservations	5	Medium	Sahil, Kritika, Sabilullah Hussaini	Done	• Create user dashboard interface • Display current and past reservations • Implement rental history tracking • Add reservation management features (view/cancel) • Design profile section for user information • PO: Define dashboard layout & user experience requirements
UI-3	Vehicle Details Page	5	High	Ayesha, Shalin, Sabilullah Hussaini	Done	• Create detailed vehicle page layout • Implement image gallery component • Display full specifications and features • Show total cost calculation • Add "Reserve Now" button • PO: Define content requirements & user flow

UI-4	Filter Search by Category/Price	5	Medium	Sahil, Moosa, Sabilullah Hussaini	Done	• Implement price range slider (\$20–\$200) • Add multiple category selection • Create real-time filter updates • Design filter sidebar component • Add clear/reset filters functionality • PO: Specify filter criteria & user experience
RES-1	Shopping Cart Functionality	3	High	Mahad, Moosa Ayesha, Sabilullah Hussaini	Done	• Create shopping cart component • Implement add/remove vehicle from cart • Design cart summary sidebar • Add cart persistence across sessions • Connect cart to reservation flow • PO: Define cart behavior & validation rules

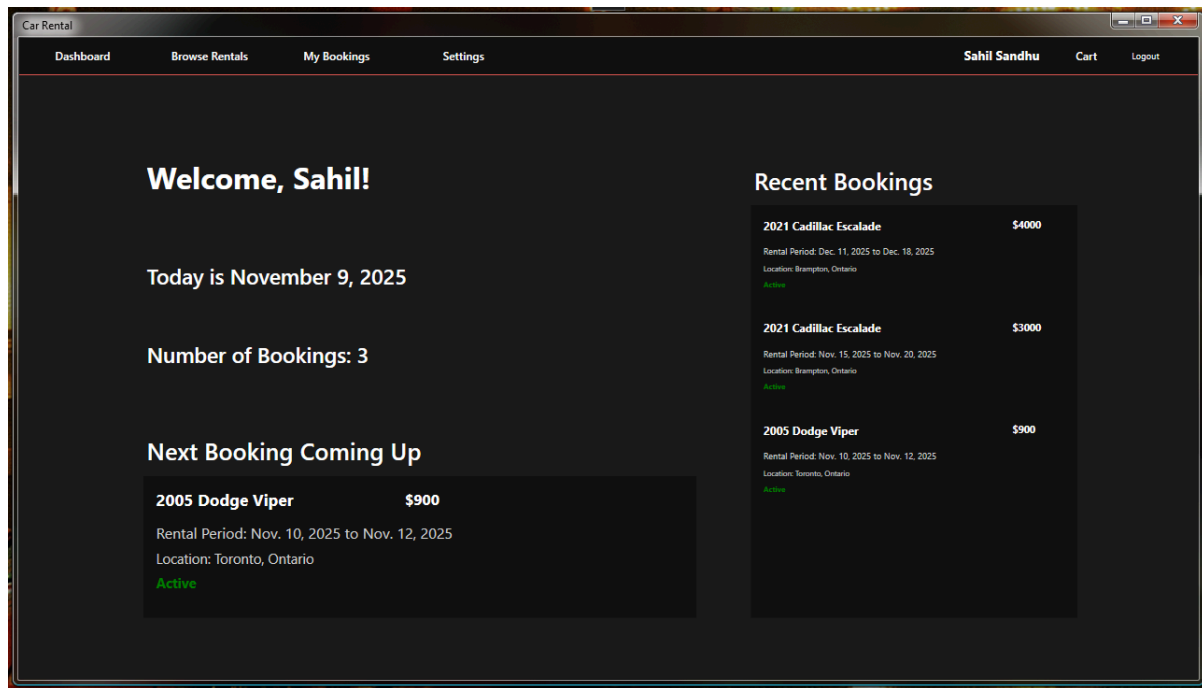
RES-2	Complete Reservation & Payment	5	High	Mahad, Moosa, Sabilullah Hussaini	Done	• Design payment form with card details input • Implement payment validation (card number, expiry, CVV) • Create booking confirmation system • Integrate with Booking entity and database • Add payment success/failure handling • PO: Define payment flow & security requirements
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Code Snippet Example & Screenshots

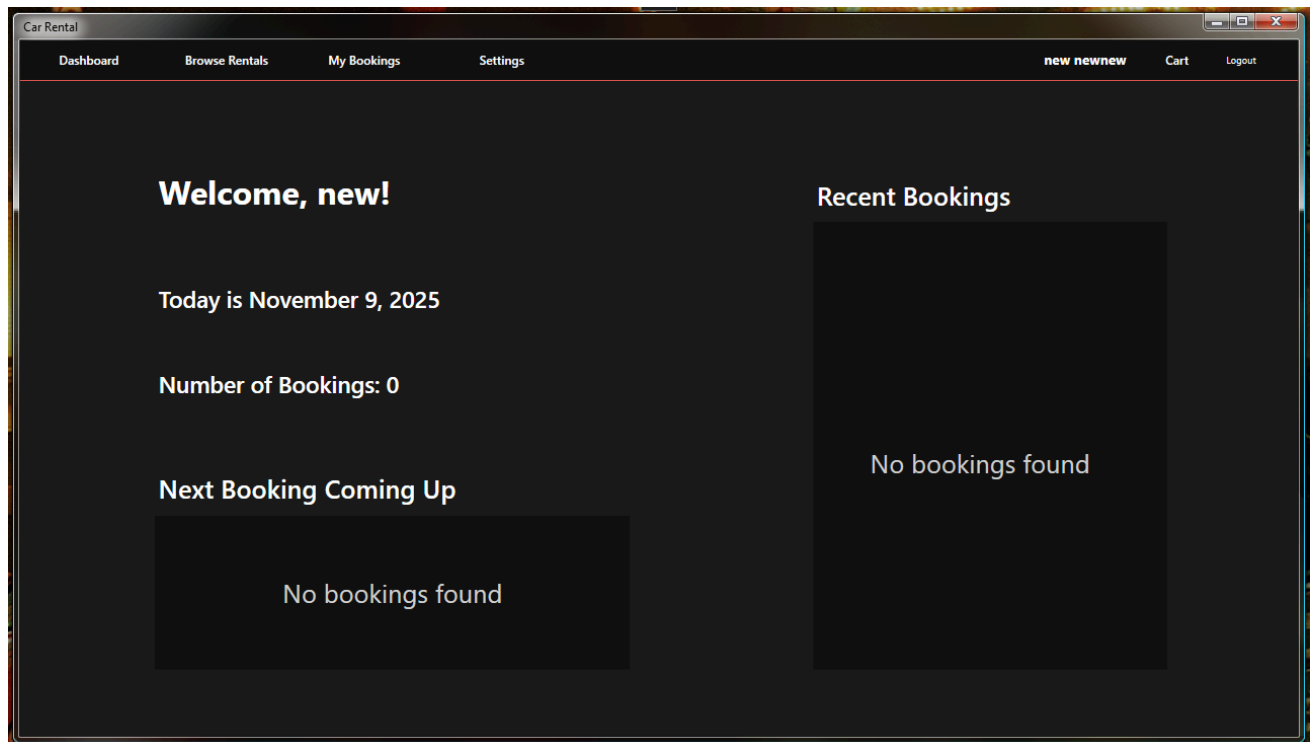
Daily Price Slider & Vehicle Type Search Filters:



Dashboard:



The dashboard shows the closest upcoming booking, as well as 4 recent bookings.

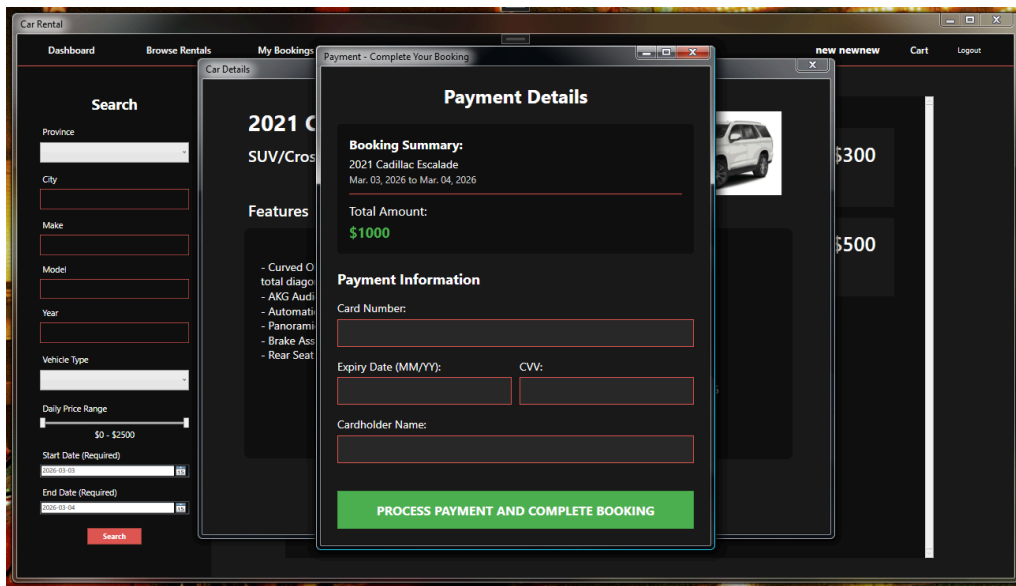


Vehicle Details Window:



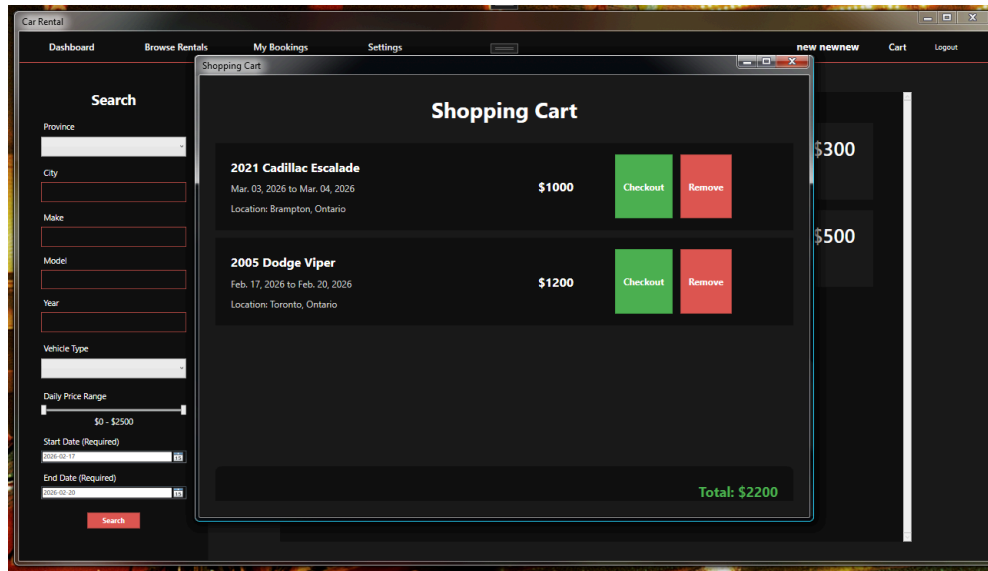
Listings now have their own details window that shows features, the vehicle type, and a summary of the booking. Users can either add a booking to the car or book instantly from this window.

Simulated Payment:



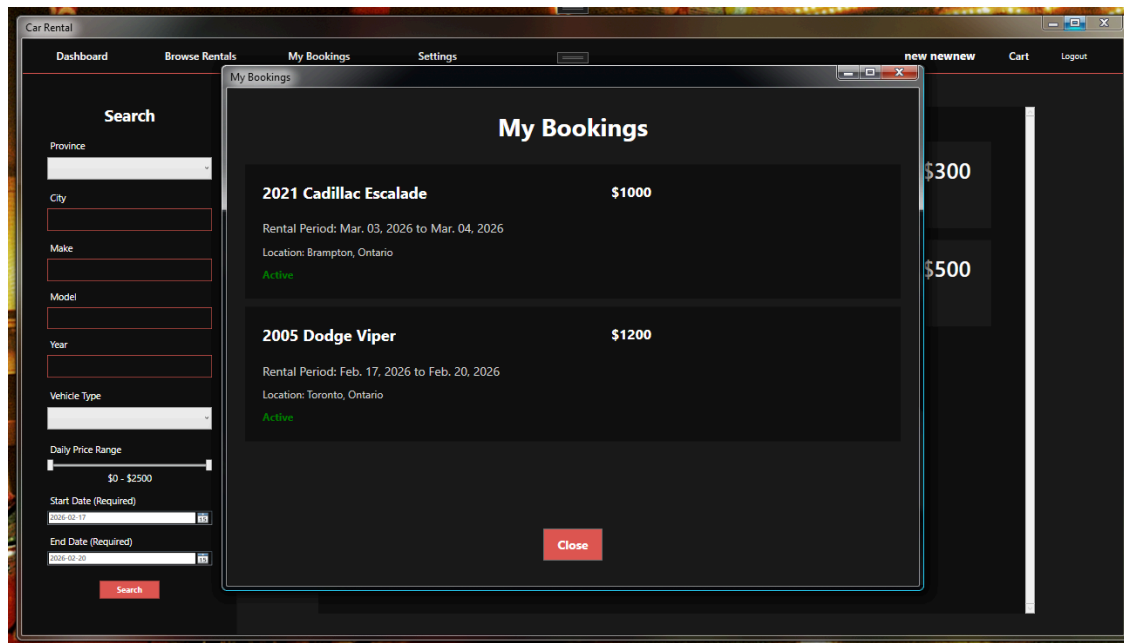
Users can enter their billing information (simulated) and finalize their booking.

Shopping Cart:



Each user has their own shopping cart. The shopping cart is persistent and saves after logging out. Each booking can be checked out individually.

My Bookings:



Users can view the My Bookings window, which shows all of their active and completed bookings.

Implementation Progress

Before development began this sprint, the team decided that some user stories should be interchanged between Sprint 2 and Sprint 3. UI-4 was brought to Sprint 2 while ADMIN-1 and ADMIN-2 were moved to Sprint 3. This was done as Sprint 3 previously had fewer user stories than Sprint 2, and the workload of Sprint 2 may have been too much for the time provided.

Demo Video:

<https://drive.google.com/file/d/147eXAeJk30c6RRILQMYS4RZy4n81eYuq/view?usp=sharing>

Testing and Review

We verified functionality using user tests and user simulation. As the foundation for the program was already completed, development was not so linear. During development, certain functionality, such as the new search filters, were tested separately from something like the cart as they do not overlap. Features such as the cart, payment simulation, and dashboard were tested using user simulation during development as they are directly related. When we finally completed all of the functionality, we did a user simulation test to cover every added feature just to make sure everything was working, even if the features were not related.

Story ID	Story Title	Card	PASS/FAIL
USER-1	Secure Account Creation	"As a new user, I want to create an account with my email and a secure password so that I can access the platform's features."	PASS
UI-3	Vehicle Details Page	"As a customer, I want to view a detailed page for a specific car with more images, full specifications, and total cost so that I can make an informed decision."	PASS
UI-4	Search Filtering	"As a customer, I want to filter search results by car category and price range so that I can narrow down my options more effectively."	PASS
RES-1	Shopping Cart	"As a customer, I want to add a selected vehicle to a shopping cart so that I can finalize my booking without losing my selection."	PASS
RES-2	Reservation Payment & Confirmation	"As a customer, I want to complete a reservation by providing payment details (simulated) so that my booking is confirmed."	PASS